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Hardware and Software

- ❖ Hardware: internal components, internal memory (RAM, ROM), graphics card, sound card, NIC, storage, input and output devices.
- ❖ Software: applications software and system software, and examples of each.
- ❖ Analogue and digital data: characteristics, differences and the need to convert between them.

Components, Operating Systems and Types of Computer

- ❖ Main components: the role of the CPU, internal memory, input and output devices and backing storage.
- ❖ Operating systems: command line, graphical, dialogue based and gesture based interfaces.
- ❖ Types of computer: desktop computers and mobile computers (laptop, smartphone, tablet, phablet).
- ❖ Emerging technologies: artificial intelligence (AI) and extended reality (virtual and augmented).

Computer Systems:

- ❖ Computer systems are now commonplace in every part of our daily life.
- ❖ Every computer system is made up of hardware and software.

A Useful Analogy:

- ❖ The pages and the ink used on the pages are equivalent to the hardware.
- ❖ The words written on those pages are equivalent to the software.
- ❖ Without the words a book is useless; without software, computers would be of little use.

Two Categories of Hardware:

- ❖ External hardware: devices outside the main unit (covered in detail in Chapter 2).
- ❖ Internal hardware: the components found inside the computer case.

Definition:

- ❖ Hardware is the general term for the physical components that make up a typical computer system.

External Hardware:

- Input devices: keyboard, mouse, camera.
- Output devices: monitor, printer, plotter.

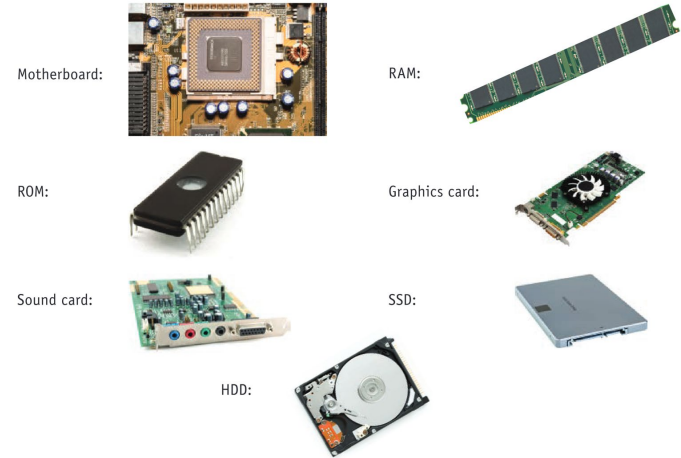
Internal Hardware:

- Motherboard and central processing unit (CPU)/processor.
- Random access memory (RAM) and read-only memory (ROM).
- Graphics card, sound card and network interface card (NIC).
- Internal storage devices: hard disk drive (HDD) and solid-state drive (SSD).

Examples of Internal Hardware

Internal Components:

- ❖ Motherboard: the main printed circuit board.
- ❖ RAM and ROM: the internal memory chips.
- ❖ Graphics card and sound card.
- ❖ HDD and SSD: internal storage devices.
- ❖ It is not always possible to see these devices inside the computer case.



▲ Figure 1.2 Examples of internal hardware

What It Is:

- ❖ A printed circuit board found in all computers.
- ❖ Allows the processor and other computer hardware to function and communicate with each other.

Major Function:

- ❖ Acts as a kind of 'hub' which other computer devices connect to.

Construction:

- ❖ Consists of a sheet of non-conductive material, such as hard plastic.
- ❖ Thin layers of copper or aluminium are printed onto this sheet to form the circuits.
- ❖ Contains several sockets and slots to connect the other components.

Central Processing Unit (CPU)

Definition:

- ❖ An electronic circuit board in a computer that can execute instructions from a computer program.
- ❖ Also called the processor; it is normally part of the computer motherboard.

Two Main Components:

- Arithmetic and logic unit (ALU): where arithmetic and logical operations are carried out.
- Control unit (CU): takes instructions, then decodes and executes the instructions.

Modern CPUs:

- ❖ Now a single integrated circuit known as a microprocessor.
- ❖ Found at the heart of most PCs and in many household devices needing control or monitoring.

Random Access Memory (RAM)

Characteristics:

- ❖ An internal chip where data is temporarily stored when running applications.
- ❖ This memory can be written to and read from.
- ❖ Its contents are lost when power to the computer is turned off.
- ❖ It is therefore referred to as a volatile or temporary memory.

What RAM Stores:

- ❖ The data, files or part of the operating system currently in use.

Note:

- ❖ RAM can be increased in size to improve the operational speed of a computer.

Read-Only Memory (ROM)

Characteristics:

- ❖ A memory used to store information that needs to be permanent.
- ❖ ROM chips cannot be altered and can only be read from, hence their name.
- ❖ Information stored on a ROM chip is not lost even when power is turned off.
- ❖ It is therefore referred to as a non-volatile memory.

What ROM Stores:

- ❖ Configuration data for a computer system.
- ❖ The boot file, often referred to as the BIOS (basic input/output system).

Graphics Card, Sound Card and NIC

Graphics Card:

- Allows the computer to send graphical information to a video display device such as a monitor, television or projector.
- Made up of a processing unit, a memory unit (usually RAM), a cooling mechanism and connections to a display unit.

Sound Card:

- An integrated circuit board that provides a computer with the ability to produce sounds.
- Allows a user to record sound input from a microphone and manipulate sound stored on a disk.

Network Interface Card (NIC):

- Allows a computer or other device, such as a printer, to be connected to a network; it can be wired or wireless.
- Each NIC is hard-coded with a unique MAC (media access control) address.

Internal HDD and SSD

Hard Disk Drive (HDD):

- ❖ Magnetic in nature.
- ❖ One of the main methods for storing data and files such as text, photos and music.
- ❖ Stores most of the system and applications software.

Solid-State Drive (SSD):

- ❖ A newer storage system used by more modern computers and all tablets.
- ❖ SSDs are replacing HDDs in many cases.
- ❖ Their function is the same as an HDD.

Definition:

- ❖ Software is the general term used for the programs that control the computer system and process data.

Applications Software:

- ❖ Provides the services that the user requires to solve a given task.
 - Word processing, spreadsheet, database (management system), control and measurement software.
 - Apps and applets, video editing, graphics editing, audio editing, computer-aided design (CAD).

System Software:

- ❖ Designed to provide a platform on which all other software can run.
 - Compilers, linkers, device drivers, operating systems and utilities.

Word Processing Software:

- ❖ Used to manipulate a text document, such as an essay or a report.
 - Creating, editing, saving and manipulating text; copy and paste functions.
 - Spell checkers and thesaurus.
 - Import photos/images into a structured page format; translation into a foreign language.

Spreadsheet Software:

- ❖ Used to organise and manipulate numerical data on a grid of lettered columns and numbered rows.
 - Cells are identified by a unique combination of column and row, for example B6.
 - Use of formulae to carry out calculations; ability to produce graphs; modelling and 'what if' calculations.

Database (Management System):

- ❖ Used to organise, manipulate and analyse data.
- ❖ Made up of one or more tables of rows (each row is a 'record') and columns (each column is a 'field').
 - Ability to carry out queries on database data and produce a report (DBMS).
 - Add, delete and modify data in a table.

Control and Measurement Software:

- ❖ Allows a computer or microprocessor to interface with sensors.
 - Measure physical quantities in the real world, such as temperature.
 - Control applications by comparing sensor data with stored data and sending out signals to alter process parameters.

Apps, Applets and Video Editing

Applets:

- ❖ Small applications that perform a single task on a device.
- ❖ Usually embedded in an HTML page on a website and executed from within a browser.

Apps:

- ❖ Software which can perform a fairly substantial task, such as video and music streaming, banking or social media.
- ❖ Originally referred to software that ran on a smartphone and could be downloaded from an 'app store'.

Video Editing Software:

- Rearranging, adding and/or removing sections of video clips and/or audio clips.
- Applying colour correction, filters and other video enhancements.
- Creating transitions between clips in the video footage.

Graphics Editing Software:

- ❖ Allows bitmap and vector images to be changed.
 - Bitmap editors change the pixels, which contain information about image brightness and colour.
 - Vector editors do not use pixels; they manipulate lines, curves and text.

Audio Editing Software:

- Alter length, start/stop time and volume of a track, and fade in/out.
- Convert between audio file formats, combine multiple sound tracks and apply noise reduction.

Computer-Aided Design (CAD):

- Produces 2D or 3D diagrams that can be rotated to view the drawing from any angle.
- Produces full dimensions, estimates manufacturing costs and predicts structural problems.

Compiler:

- ❖ Translates a program written in a high-level language (HLL) into machine code understood by the computer.
- ❖ The original program is called the source code; the code after compilation is called the object code.
- ❖ Once compiled, the machine code can be used again and again without recompilation.
- ❖ Examples of high-level languages: Java, Python, Visual Basic, Fortran, C++ and Algol.

Linker:

- ❖ Takes one or more object files produced by a compiler and combines them into a single program.
- ❖ Programmers can write separate pieces of code called modules, which simplifies the programming task.
- ❖ The linker puts all the modules together to form the final program.

Drivers, Operating Systems and Utilities

Device Driver:

- ❖ Software that enables one or more hardware devices to communicate with the computer's operating system.
- ❖ Without drivers, a hardware device such as a printer would be unable to work with the computer.
- ❖ As soon as a device is plugged into a USB port, the operating system looks for the appropriate driver.

Operating System (OS):

- ❖ Software running in the background of a computer system that manages many of the basic functions.
- ❖ Allows input/output operations, user communication, error handling, loading and running of programs, and security.

Utilities:

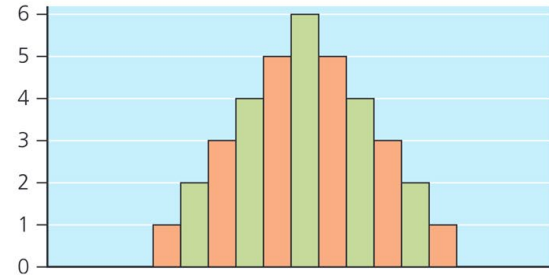
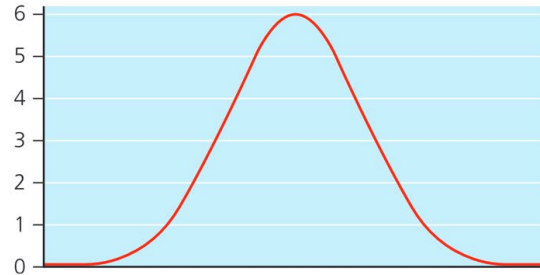
- ❖ Programs that help to manage, maintain and control computer resources.
 - Antivirus, anti-spyware, backup of files, disk repair, file management, security, screensavers, disk defragmenter.

Digital Data:

- ❖ Computers can only understand data in a binary format: a base 2 number system using only 0 and 1.
- ❖ Digital data can only have discrete, discontinuous values.

Analogue Data:

- ❖ Data in the real world is analogue in nature.
- ❖ Analogue data is physical data that changes smoothly from one value to the next.
- ❖ An infinite number of values can exist between any two points, because the curve is smooth in shape.



▲ Figure 1.5 Analogue and digital data

Converting Analogue and Digital Data

Analogue to Digital Converter (ADC):

- ❖ If analogue data is being sent to a computer, it must first be converted into digital data.
- ❖ This is done by hardware known as an analogue to digital converter (ADC).

Digital to Analogue Converter (DAC):

- ❖ If the computer is controlling a device, such as a motor, the digital output must be converted into analogue form.
- ❖ This is done by hardware known as a digital to analogue converter (DAC).

Why Conversion Is Needed:

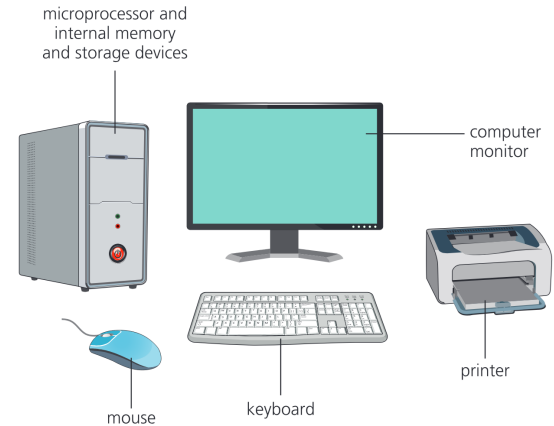
- ❖ Computers can only process digital data.
- ❖ Real-world data and the devices a computer controls are analogue.

A Typical Computer System:

- ❖ Made up of hardware and software.
- ❖ Consists of input devices, output devices and secondary storage.

Four Key Internal Components:

- The central processing unit (CPU), contained on the motherboard.
- Internal hard disk drive or solid-state drive.
- Random access memory (RAM).
- Read-only memory (ROM).



▲ **Figure 1.6** A typical computer system

Table 1.1 Examples of input, output and secondary storage devices

Device	Examples
Input devices	keyboard, mouse, camera, microphone, sensor, scanner
Output devices	monitor, printer, speakers, projector, (graph) plotter
Secondary storage devices	hard disk drive, solid-state drive, pen drive

Role:

- ❖ Interprets and executes the commands from the computer hardware and software.
- ❖ Processes the instructions entered into the computer in order to produce an output.
- ❖ It is normally part of the computer motherboard.

Development:

- ❖ CPUs used to be made up of discrete components and numerous small integrated circuits.
- ❖ Modern manufacturing techniques mean the CPU is now a single integrated circuit: a microprocessor.

The CPU Is Made Up Of:

- A control unit, which controls the input and output devices.
- An arithmetic and logic unit (ALU), which carries out calculations and makes logical decisions.
- Small memory locations called registers.

The Boot File:

- ❖ ROM also contains some coding known as the boot file.
- ❖ This code tells the computer what to do when it first starts up.
- ❖ It is often referred to as the BIOS (basic input/output system).

What Happens at Start-Up:

- The BIOS carries out a hardware check to find out if all the devices are present and functional.
- It then loads the operating system into the RAM.

CMOS:

- ❖ The BIOS stores the date, time and system configuration in a non-volatile chip called a CMOS.
- ❖ CMOS stands for complementary metal oxide semiconductor; it is usually battery powered.

Comparison of RAM and ROM

Table 1.2 RAM and ROM differences

RAM	ROM
Temporary memory device	Permanent memory device
Volatile memory	Non-volatile memory device
Can be written to and read from	Read-only; data stored cannot be altered
Used to store data, files, programs and part of the operating system (OS) currently in use	Used to store the BIOS and other data needed at start up
Can be increased in size to improve the operational speed of a computer	

Input Devices:

- ❖ Hardware that allows data to be entered into a computer.
- ❖ Use manual entry, such as a keyboard or a mouse.
- ❖ Or use direct data entry, such as sensors or optical character readers.
- ❖ They turn input into a form the computer can understand.

Output Devices:

- ❖ Show the computer's output in a form that is understood by a human.
- ❖ For example, as text on a printer or moving images on a monitor.

Devices That Do Both:

- ❖ A touch screen or a DVD writer/player can act as both an input and an output device.

Comparison of Input and Output Devices

Table 1.3 Comparison of input and output devices

Input devices	Output devices
An input device is any hardware device that allows a user to enter data or instructions into a computer directly.	An output device is any hardware device that takes the output data from a computer and puts it into a human-readable format or uses it to control another device.
An input device can send data to another device, but it cannot receive data from another device.	An output device is capable of receiving data from another device in order to generate an output, but it cannot send data to another device.
Input devices are necessary for a computer to receive commands and data to process; they are under the control of the user or can be direct data entry.	Output devices are needed by a computer so it can share the results of its processing with a human; they are under the control of the computer.
Input devices can be fairly complicated, because they have to ensure that the user can interact with the computer correctly.	Output devices are less complex than input devices, because they only have to turn computer signals into an output.

Purpose:

- ❖ Used to permanently store large amounts of data.
- ❖ This normally takes the form of the internal hard disk drive (HDD) or solid-state drive (SSD).
- ❖ It stores the applications software, disk operating system and files such as text, photos and music.

Key Feature:

- ❖ It must store data permanently, that is, it must be non-volatile.
- ❖ It is not directly addressable: it cannot be read directly by the CPU.

Fixed or Removable:

- ❖ Fixed backing storage is very often internal to the computer.
- ❖ Removable backing storage can be used as a backup in case of data loss or corruption.
- ❖ Examples of removable storage: external HDD, external SSD and Blu-ray discs.

Internal Memory vs Backing Storage

Table 1.4 Comparison of internal memory and backing storage

Internal memory	Backing storage
RAM contents are lost when the computer is powered down; ROM contents are readable only.	Backing storage devices hold their contents permanently, even when powered down.
RAM and ROM are much smaller memories than backing storage.	Has a considerably larger capacity to store data than RAM or ROM.
Data access time on RAM and ROM is extremely fast.	Has a much slower data access time than RAM and ROM.
Much more expensive per byte than backing storage devices.	Is much cheaper per byte than RAM or ROM.
RAM and ROM are fixed inside the computer (internal memories).	Can either be fixed (external or internal) or it can be removable.
RAM and ROM can be read directly by the CPU.	Data must first be moved into RAM before the CPU can read it, so it is not directly addressable.

Definition:

- ❖ Special software developed to enable computer systems to function and to allow users to communicate with them.

General Tasks of an Operating System:

- Control of the operation of the input, output and backing storage devices.
- Supervising the loading, running and storage of applications programs.
- Dealing with errors that occur in application programs.
- Maintaining security of the whole computer system.
- Maintaining a computer log, which details computer usage.
- Allowing communication between the user and the computer system through a user interface.

Operating Systems Offer Four Types of User Interface:

- Command line interface (CLI).
- Graphical user interface (GUI).
- Dialogue-based user interface.
- Gesture-based user interface.

Why the Interface Matters:

- ❖ The interface determines how the user communicates with the computer.
- ❖ Without an operating system and its interface, most computers would be very user-unfriendly.
- ❖ The majority of users would find it almost impossible to work with computers day-to-day.

Command Line Interface (CLI)

Characteristics:

- ❖ Requires a user to type in instructions to choose options from menus, open software and so on.
- ❖ A number of commands often need to be typed in, for example to save or load a file.
- ❖ The user has to learn a number of commands just to carry out basic operations.

Speed:

- ❖ It is slow, because these commands must be keyed in every time an operation is carried out.

Main Benefit:

- ❖ The user is in direct communication with the computer.
- ❖ The user is not restricted to a number of pre-determined options.

Example:

- ❖ Windows has a CLI called the 'command prompt'.

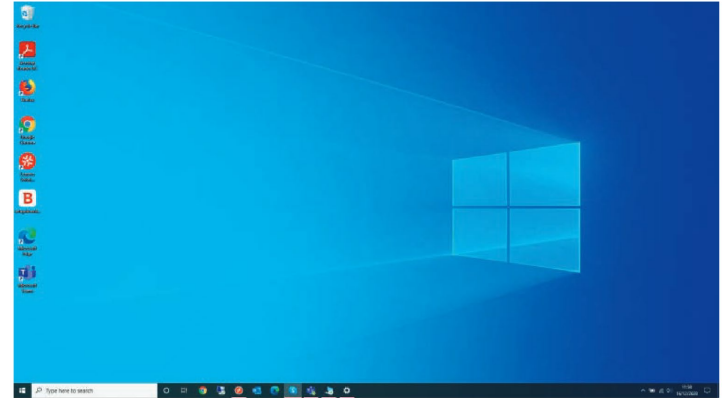
Graphical User Interface (GUI)

Characteristics:

- ❖ Allows the user to interact with a computer using pictures or symbols (icons).
- ❖ Removes the need to type in a number of commands.

WIMP:

- ❖ Windows, icons, menu and pointing device.
- ❖ Developed for use on personal computers (PC).
- ❖ A mouse controls a cursor and icons are selected to open/run windows.



▲ Figure 1.8 Screen image showing icons

Windows:

- ❖ Each window contains an application.
- ❖ Modern computer systems allow several windows to be open at the same time.
- ❖ A window manager looks after the interaction between windows, the applications and the window system.
- ❖ The window system handles the pointing devices and the cursor's position.

Post-WIMP Interaction:

- ❖ Used by devices such as touch screen smartphones and tablets.
- ❖ Fingers are in contact with the screen, allowing actions such as pinching and rotating.
- ❖ These actions would be difficult to do using a single pointer and device such as a mouse.

CLI and GUI: Advantages and Disadvantages

Table 1.5 Advantages and disadvantages of CLI and GUI interfaces

Interface	Advantages	Disadvantages
Command line interface (CLI)	The user is in direct communication with the computer. The user is not restricted to a number of pre-determined options. It is possible to alter computer configuration settings.	The user needs to learn a number of commands to carry out basic operations. All commands need to be typed in, which takes time and can be error-prone. Each command must be typed in using the correct format and spelling.
Graphical user interface (GUI)	The user does not need to learn any commands. It is more user-friendly; icons are used to represent applications. A pointing device is used to click on an icon to launch the application.	This type of interface uses up considerably more computer memory than a CLI. The user is limited to the icons provided on the screen. Needs a more complex operating system, which can be slower to execute commands.

Who Would Use Each Interface?

Command Line Interface (CLI):

- ❖ A programmer, analyst or technician.
- ❖ Somebody who needs direct communication with a computer to develop new software.
- ❖ Somebody who needs to locate errors and remove them.
- ❖ Somebody who needs to initiate memory dumps: the contents of computer memory at some moment in time.

Graphical User Interface (GUI):

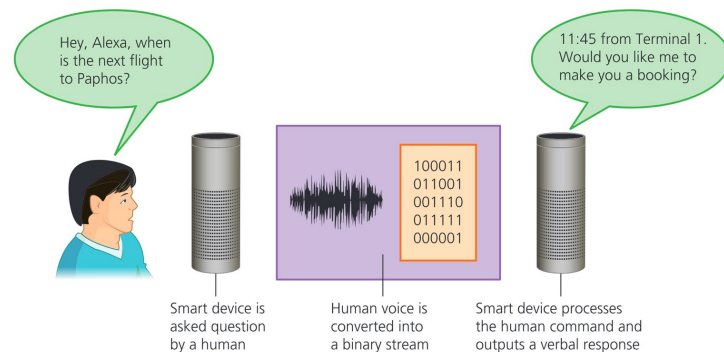
- ❖ The end-user, who does not need any great knowledge of how the computer works.
- ❖ A person who uses the computer to run software, play games or store/manipulate photographs.

Characteristics:

- ❖ Use the human voice to give commands to a computer system.
- ❖ Natural speech enables the user to interact intuitively with the device.

Examples of Use:

- Luxury modern cars: voice activation controls the entertainment or satellite navigation system.
- The home: voice commands switch lights on/off and operate electronic equipment.
- Amazon Alexa, Google Now, Apple Siri and Microsoft Cortana act as personal assistants.



▲ Figure 1.9 Smart voice activated devices

Characteristics:

- ❖ Rely on human interaction by the moving of hands, head or even the feet.
- ❖ Allow humans to interface with a computer in a more natural fashion, without mechanical devices.
- ❖ Use techniques known as computer vision and image processing.

Examples in a Car:

- Rotating a finger clockwise near the radio increases the sound volume; anti-clockwise reduces it.
- Opening the thumb and next finger changes the track being listened to.
- Moving the foot under the rear bumper automatically opens the boot lid.
- Moving a hand near a window switch automatically opens a window.

How It Works:

- ❖ A sensor or camera picks up the gesture and sends a signal to an on-board computer.

Dialogue and Gesture Interfaces Compared

Table 1.6 Advantages and disadvantages of dialogue and gesture interfaces

Interface	Advantages	Disadvantages
Dialogue-based interface	No need for a driver to take their hands off the steering wheel. Very useful in the home for people with disabilities, as tasks are done by the spoken word only. Possible to use as a security feature, because voice recognition could identify a person.	Still unreliable, with many commands not being recognised or needing to be repeated several times, especially with background noise. Can be quite complex to set up. The user needs to know which commands can be used.
Gesture-based interface	Replaces mechanical input devices. No physical contact required. Very natural interface for a human operator. No training needed to interface with the computer.	Possible for unintentional movement to be picked up. Only works fairly near to the camera or sensor (maximum of 1.5 metres). May only accept a limited number of movements.

Desktop Computers

Definition:

- ❖ A general-purpose computer made up of a separate monitor, keyboard, mouse and processor unit.
- ❖ Distinguished from a laptop by being made up of a number of separate components.
- ❖ These separate components make desktop computers not very portable.



▲ Figure 1.10 Desktop computer

Advantages of Desktop Computers

- ❖ Spare parts and connections tend to be standardised, which usually results in lower costs.
- ❖ Easier, and less expensive, to upgrade or expand.
- ❖ Tend to have a better specification for a given price, for example a faster processor.
- ❖ Power consumption is not critical, because they usually plug straight into a wall socket.
- ❖ The larger casings allow a better dissipation of any heat build-up.
- ❖ Less likelihood of being damaged or stolen, because they are usually fixed in one location.
- ❖ Internet access can be more stable, because a wired connection gives the same data transfer rate.
- ❖ Laptops usually use wireless connectivity, where the signal can be very variable.

Disadvantages of Desktop Computers

- ❖ They are not particularly portable, because they are made up of separate components.
- ❖ They tend to be more complicated, because all the components need to be hooked up by wiring.
- ❖ The wiring also clutters up the desk space.
- ❖ Files must be copied onto a memory stick to do work elsewhere, such as office work at home.
- ❖ Cloud storage has diminished this disadvantage recently.
- ❖ However, it still may not be possible to save sensitive data files on the cloud.

Uses of Desktop Computers

- ❖ Office and business work.
 - Word processing, spreadsheets, finance software and databases are the main uses.
- ❖ Educational use.
 - Using interactive software to teach or learn from.
- ❖ Use as a gaming device.
 - For example, games such as chess, crossword puzzles and fantasy games.
- ❖ General entertainment.
 - For example, live or 'catch-up' streaming of television programmes.

Definition:

- ❖ A group of computers which are considerably more portable than desktop computers.

Four Categories:

- Laptop computers.
- Smartphones.
- Tablets.
- Phablets.

Definition:

- ❖ The monitor, keyboard, pointing device and processor are all together in one single unit.
- ❖ Also referred to as a notebook; this makes them extremely portable.

Key Features:

- Lightweight, to aid portability.
- Low power consumption and long battery life.
- Low heat output, because cooling is very important.



▲ Figure 1.11 Laptop computer

Laptops: Advantages and Disadvantages

Advantages Compared to Desktop Computers:

- Portability: the monitor, pointing device, keyboard, processor and backing store are one single unit.
- No trailing wires, because only one single cord is used.
- They take up much less room on a desk, so they can be used anywhere, for example in a cafe.
- Their portability allows them to take full advantage of Wi-Fi features.
- Because they are portable, they can link into any multimedia system.

Disadvantages Compared to Desktop Computers:

- Because they are easily portable, they are also easier to steal.
- They have limited battery life, so the user may need to carry a heavy power adaptor.
- The keyboards and pointing devices can sometimes be more awkward to use.
- It is not always possible to upgrade them, for instance by adding more RAM.

Uses of Laptop Computers

- ❖ Office and business work: word processing, spreadsheets, finance software and databases.
- ❖ Educational use: using interactive software to teach or learn from.
- ❖ Use as a gaming device: for example chess, crossword puzzles and fantasy games.
- ❖ General entertainment: for example live or 'catch-up' streaming of television programmes.
- ❖ Control and monitoring.
 - They can be operated anywhere using their internal battery.
 - It is possible to use laptops to gather data directly from the environment.
 - For example, they can be plugged into a gas boiler during maintenance to monitor its performance.

Definition:

- ❖ Allow normal phone calls to be made.
- ❖ Have an operating system, such as iOS, Android or Windows.
- ❖ Run a number of computer applications, known as apps or applets.

What They Allow:

- Send and receive emails, and use a number of apps.
- Use a camera feature to take photos or videos.
- Act as MP3/4 players for music and videos.

Internet Connection:

- ❖ Wi-Fi hot spots or 3G/4G/5G mobile phone networks.



▲ Figure 1.12 Smartphone

Typical Features of Smartphones

- ❖ High-definition, anti-glare displays.
- ❖ Front- and back-facing cameras.
 - Used to take photos and videos, or act as a webcam for video calls over the internet.
- ❖ Lower weight and longer battery life than laptops.
- ❖ Use Bluetooth for connection to printers and other devices.
- ❖ Make use of flash (solid state) memory and cloud storage facilities.
 - Used to back up and synchronise (often just referred to as 'sync') data sources.
- ❖ New generations use OLED (organic light emitting diode) touch screens, making them thinner and lighter.

Smartphone Sensors and Future Technology

Sensors:

- Proximity sensors detect if the device is close to the ear, blocking unintended 'touches'.
- Accelerometers detect movement and orientation, moving the display from portrait to landscape.
- Accelerometers also allow a phone to be used as a 'steering wheel' in car racing games.
- Speech recognition systems, such as Siri, let the user ask the device to look for things.

Future Technology: Li-Fi

- OLED screens coated with a crystalline layer allow phones to be partially solar powered.
- Li-Fi communication uses visible light rather than radio waves.
- Considerably faster than Wi-Fi and more secure, by stopping internet 'piggybacking'.
- Can be used during flights, because it does not interfere with flight control systems.

Advantages of Smartphones

- ❖ Very small in size and lightweight, so very easy to carry and have on your person at all times.
 - This is more difficult with laptops, because they are much bulkier and much heavier.
- ❖ You can make phone calls, but also connect to the internet while on the move.
- ❖ They can be used almost anywhere, because they use Wi-Fi and mobile phone networks.
- ❖ They have apps which make use of sensor data provided by the smartphone.
 - For instance location data for maps, providing services not available on desktops or laptops.
- ❖ They have a reasonable battery life compared to laptops.

Disadvantages of Smartphones

- ❖ The small screens and keyboards make pages difficult to read.
- ❖ It is more difficult and slower when typing things in.
- ❖ Web browsing and photography can quickly drain the battery.
- ❖ Memory size in most phones is not very large compared to laptops and desktops.
 - It is comparable with tablets; the latest generation come with 1 TiB memories.
- ❖ Not all website features are compatible with smartphone operating systems.
- ❖ Because of their small size, it is much easier to lose a smartphone or for it to be stolen.
- ❖ The data transfer rate using mobile phone networks can be slower than with Wi-Fi.

Advantages of Tablets Compared to Laptops

- ❖ Very fast to switch on, with no time delay waiting for the operating system to load up.
- ❖ Fully portable: they are so lightweight that they can be carried anywhere.
- ❖ Touch screen technology means they are simple to use and need no other input devices.
- ❖ Can use several apps as standard, such as a built-in camera and MP3/4 players.
- ❖ Not much heat, because they use solid-state technology.
- ❖ The battery life of a tablet is a lot longer.
- ❖ Pressing the power button puts it into standby but keeps it connected to the internet.
 - The user still hears alerts when emails or other 'events' are received.

Disadvantages of Tablets Compared to Laptops

- ❖ They tend to be rather expensive when compared to laptops.
 - This will probably change with time as they become more common.
- ❖ They often have limited memory or storage when compared to a laptop.
 - Some of the latest devices have 1 TiB memory capacity.
- ❖ If 3G/4G/5G networks are used, they can be expensive to run when the internet is accessed frequently.
- ❖ Typing on a touch screen can be slow and error-prone compared to a standard keyboard.
- ❖ Transferring files often has to be done through an 'application store'.
 - This lack of 'drag and drop' facility can prove to be irritating to users.
- ❖ Laptops support more file formats and are better equipped to run different types of software.

Definition:

- ❖ A hybrid between a tablet and a smartphone.
- ❖ Much larger screens than a smartphone, but smaller than a tablet.
- ❖ All the features of a smartphone apply, together with the typical features of a tablet.

Size Comparison:

- Smartphones: up to 5.1 inches in size (that is, 13 cm).
- Phablets: between 5.1 inches and 7 inches (that is, 13 cm to 18 cm).
- Tablets: over 7 inches in size (that is, over 18 cm).



▲ Figure 1.14 Comparison of smartphone, phablet and tablet

Uses of Mobile Devices

- ❖ Entertainment: streaming of music, videos and television programmes.
- ❖ Gaming, including group games.
- ❖ As a camera or video camera; the quality now matches a good digital camera.
- ❖ Internet use: online sales, social networks and using QR codes.
- ❖ Sending and receiving emails.
- ❖ Global positioning system: use of maps to navigate to a location.
- ❖ Calendar functions.

Uses of Mobile Devices cont.

- ❖ Telephone banking: sending and receiving money using banking apps.
- ❖ Voice over Internet Protocol (VoIP): a telephone network using the internet, which allows video calling.
- ❖ Instant access to social networks: social contact with friends anywhere in the world.
- ❖ Instant messaging.
- ❖ Office and business management, particularly rapid voice and video communication.
- ❖ Education use: using interactive software to teach or learn from.
- ❖ Remotely controlled devices.
 - Home devices such as microwave ovens contain embedded microprocessors.
 - An internet-enabled smartphone or tablet can start/stop the oven from several kilometres away.

Definition:

- ❖ A machine or application which carries out a task that requires some degree of intelligence.

Examples:

- The use of language.
- Recognising a person's face.
- The ability to operate machinery, such as a car, aeroplane or train.
- Analysing data to predict the outcome of a future event, for example weather forecasting.

What AI Does:

- ❖ Duplicates human tasks which require decision-making and problem-solving skills.
- ❖ Many tasks presently done by humans will eventually be replaced by robots or computers.

Isaac Asimov's Three Laws of Robotics:

- Robots may not injure a human through action or inaction.
- Robots must obey orders given by humans without question.
- A robot must protect itself unless it conflicts with the two laws above.



▲ **Figure 1.15** An autonomous (driverless) vehicle – we already have driverless trains and autopilots on aeroplanes, but future developments include driverless cars.



▲ **Figure 1.16** Robotic research is leading to improvements in technology to help amputees and people with disabilities.



▲ **Figure 1.17** Robots are used to help people carry out dangerous or unpleasant tasks – for example, bomb disposal, welding of car bodies, entering nuclear disaster areas (such as Chernobyl or Fukushima) where the radiation would kill a human in under two minutes.

Positive and Negative Impacts of AI

Positive Impacts:

- ❖ Improvements in safety and quality of services and products.
- ❖ Autonomous (driverless) vehicles: driverless trains and aeroplane autopilots already exist.
- ❖ Robotic research is improving technology to help amputees and people with disabilities.
- ❖ Robots help people carry out dangerous or unpleasant tasks.
 - For example bomb disposal, welding of car bodies and entering nuclear disaster areas.

Negative Impacts:

- ❖ Could lead to many job losses, although new technical jobs would also be created.
 - Jobs could be lost in manufacturing, and for bus, taxi, lorry and train drivers.
- ❖ Dependency on technology and the inability to carry out tasks done by robots.
- ❖ Loss of skills, as humans have been replaced by machines and software applications.

Definition:

- ❖ Refers to real and virtual combined environments.
- ❖ A 'catch all' term for all immersive technologies.

The Three Most Common Examples:

- Augmented reality (AR).
- Virtual reality (VR).
- Mixed reality (MR).

Purpose:

- ❖ Extend the reality we experience by blending the virtual and real worlds.
- ❖ Or by creating a fully immersive experience.

Features:

- ❖ Allows the user to experience the relationship between digital (virtual) and physical (real) worlds.
- ❖ Virtual information and objects are overlaid onto real-world situations.
- ❖ The real world is enhanced with digital details, such as images, text and animation.
- ❖ The user experiences the AR world through special goggles or via smartphone/phablet screens.
- ❖ The user is not isolated from the real world and can still see what is going on in front of them.

Example:

- ❖ The Pokemon GO game, which overlays digital creatures onto real-world situations.

The Future Impact of Augmented Reality

- ❖ Safety and rescue operations.
 - 3D images of an area let a rescue team interact with the environment and try out procedures first.
- ❖ Entertainment.
 - AR takes users into a virtual environment where it is possible to interact with the characters.
- ❖ Shopping and retail: one of the big areas.
 - Using a smartphone camera to try out make-up and see how it looks before buying it.
 - A virtual tour of a new car, where you can 'sit' in the interior and try the driving experience.
- ❖ Healthcare.
 - Doctors use AR to have a better understanding of a patient's body.
 - Software such as Echopixel builds a 3D image from CT scans to help with surgery and diagnosis.

Features:

- ❖ Takes the user out of the real-world environment into a virtual (unreal) digital environment.
- ❖ In contrast to AR, the user is fully immersed in a simulated digital world.
- ❖ Users must wear a VR headset or head-mounted display, allowing a 360° view of the virtual world.
- ❖ This 'fools' the brain into believing they are walking on an ocean bed, in an alien world or inside a volcano.

Current Uses:

- ❖ Medicine, for teaching operation procedures.
- ❖ Construction, engineering and the military.

The Future Impact of Virtual Reality

- ❖ Military applications: for example, training to operate a new tank.
- ❖ Education: for example, looking inside an ancient building as part of a history lesson.
- ❖ Healthcare: for example, as a diagnostic tool to recommend treatment.
- ❖ Entertainment: games where gloves, goggles or helmets are worn to fully immerse players.
- ❖ Fashion: doing fashion shows before the real thing, seeing the clothes on people and checking the venue.
- ❖ Heritage: allowing users to walk around and close up to monuments like Stonehenge.

The Future Impact of Virtual Reality cont.

- ❖ Business: training courses and role-playing scenarios for staff.
- ❖ Engineering: seeing how new designs like bridges will look in an existing environment.
- ❖ Sport: a golfer trying to improve their swing can use this technology and get feedback to improve their game.
- ❖ Media: interactive special effects in movies.
- ❖ Scientific visualisation: for example, part of a molecular structure in chemistry, or a cell in biology.